A Tiny Swimming Robot Powered by Living Muscle

From: ScienceDaily Top Technology Mined: 2026-10-25 3 min read

Scientists at MIT have built a tiny swimming robot that moves with living muscle instead of a motor. The robot is a thin, bendy **gel** frame about the size of a stick of gum, with two fins covered in **muscle cells** much thinner than a human hair. The cells were changed so they squeeze tight when light shines on them.

Shining light on one fin makes it flap and push the robot through the water. By picking which fin gets light, scientists can steer it and change its speed. In tests, it swam and turned through a simple underwater **maze**. At its fastest, it moved about four body lengths in one minute, much slower than an Olympic swimmer's 65.

"The robot's quite strong, given its size," said study author Ritu Raman of MIT.

The study, published in Advanced Functional Materials, describes the first very thin, flat robot that moves with living muscle. Raman believes such soft robots could one day do gentle jobs in places too **fragile** for normal machines, because living **tissue** is soft and can heal itself.

Word Treasure

muscle cells -- Tiny living parts of a body that tighten to make movement.

gel -- A soft, jelly-like material that can bend easily.

tissue -- Living material that makes up parts of a body.

fragile -- Easy to break or damage, so it needs gentle care.

maze -- A path with twists and turns that you have to find your way through.

Background you need

A muscle is the soft part of a body that tightens to make movement.

In this robot, tiny muscle cells on its fins tighten when light shines on them, so the fins flap and push it through water.

Scientists hope soft robots like this can one day work gently in places where hard machines could cause harm.

Break it down

WHO: Scientists at MIT and study author Ritu Raman
WHAT: Built a tiny swimming robot that moves with living muscle instead of a motor
WHERE: In water, through a simple underwater maze
WHY: To explore soft robots that could one day do gentle jobs in places too fragile for normal machines

Why it matters

It shows kids how science can turn a tiny idea into a gentle helper for delicate jobs.

Test what you remember

1. **What moves the tiny swimming robot instead of a motor?**
   - (A) Wind
   - (B) A motor
   - (C) Living muscle
   - (D) A battery
2. **What happens to the muscle cells when light shines on them?**
   - (A) They squeeze tight
   - (B) They change color
   - (C) They stop working
   - (D) They melt
3. **How can scientists steer the robot and change its speed?**
   - (A) By picking which fin gets light
   - (B) By shouting at it
   - (C) By changing the water temperature
   - (D) By pulling it with a string
4. **About how big is the robot?**
   - (A) About the size of a house
   - (B) About the size of a whale
   - (C) About the size of a car
   - (D) About the size of a stick of gum
5. **What did the robot do in tests?**
   - (A) It swam and turned through a simple underwater maze
   - (B) It cleaned a swimming pool
   - (C) It flew through the air
   - (D) It raced an Olympic swimmer
6. **At its fastest, about how many body lengths did the robot move in one minute?**
   - (A) One hundred
   - (B) Four
   - (C) Sixty-five
   - (D) Two

Answer key (try the quiz first!): 1: C 2: A 3: A 4: D 5: A 6: B

Questions to think about

1. Ritu Raman: Believes such soft robots could one day do gentle jobs in places too fragile for normal machines, because living tissue is soft and can heal itself.

MY THOUGHTS (at least 20 words):